

Lab 3: Web API using Custom Model Class and Custom Filters

Objective

To create an ASP.NET Core Web API using a custom model class, implement GET, POST, and PUT action methods, demonstrate the use of the FromBody attribute, apply custom authorization and exception filters, and test the API using Swagger.

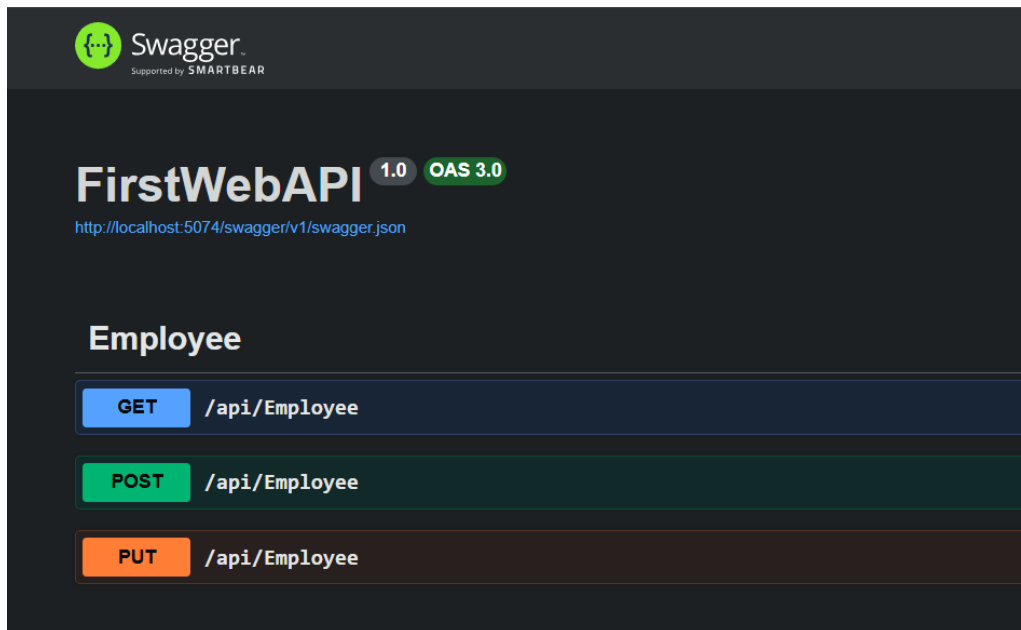
Code

Program.cs:

```
1  using FirstWebAPI.Filters;
2
3  var builder = WebApplication.CreateBuilder(args);
4
5  builder.Services.AddControllers(options =>
6  {
7      options.Filters.Add<CustomExceptionFilter>();
8  });
9
10 builder.Services.AddEndpointsApiExplorer();
11 builder.Services.AddSwaggerGen();
12
13 var app = builder.Build();
14
15 app.UseSwagger();
16 app.UseSwaggerUI();
17
18 app.UseAuthorization();
19
20 app.MapControllers();
```

Output

```
type.  
C:\Users\laksh\FirstWebAPI\Filters\CustomAuthFilter.cs(18,18): warning CS8602: Der  
uild succeeded with 7 warning(s) in 1.6s  
S C:\Users\laksh\FirstWebAPI> dotnet run  
sing launch settings from C:\Users\laksh\FirstWebAPI\Properties\launchSettings.json..  
uilding...  
nfo: Microsoft.Hosting.Lifetime[14]  
Now listening on: http://localhost:5074  
nfo: Microsoft.Hosting.Lifetime[0]  
Application started. Press Ctrl+C to shut down.  
nfo: Microsoft.Hosting.Lifetime[0]  
Hosting environment: Development  
nfo: Microsoft.Hosting.Lifetime[0]  
Content root path: C:\Users\laksh\FirstWebAPI
```



PUT /api/Employee

Parameters

No parameters

Request body

[Edit Value](#) | [Schema](#)

```
{
  "id": 1,
  "name": "Bhavya",
  "salary": 99999,
  "permanent": true,
  "department": {
    "id": 1,
    "name": "cse"
  },
  "skills": [
    {
      "id": 1,
      "name": "java"
    }
  ],
  "dateOfBirth": "2006-07-24T13:22:30.962Z"
}
```